LP $\Rightarrow$ CP 3D Printed Reflectarray

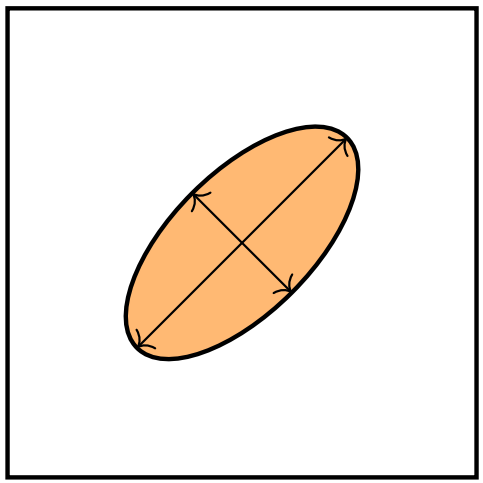
Prepared for meeting on 11.3.2026

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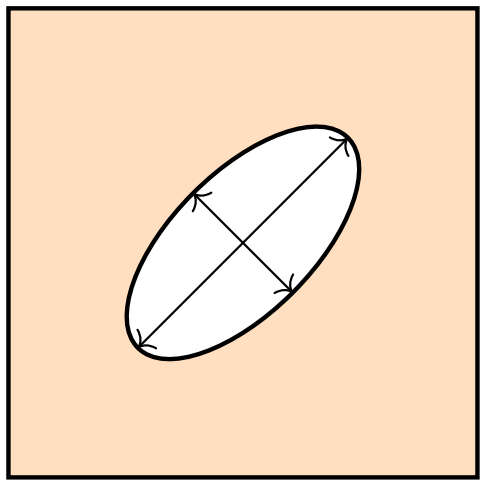
What We Are Trying To Do

- Design a **reflectarray antenna** whose unit cell is **purely dielectric** (no metallisation of the elements).
  - Only the ground plane is conductive.
  - Elements additively manufactured (**FDM** / SLA).
- Simultaneously achieve **beam focusing** *and* **LP** $\Rightarrow$ **CP polarization conversion** from a single linearly polarized feed.
- The cell must therefore control **two orthogonal reflection phases** independently.
- Target frequency: dictated purely by manufacturing technology – resolution x print bed size.

Proposed Unit Cell Geometry



elliptical **pillar**



elliptical **void**

- **Pillar** variant: better printability (no thin walls).
- **Void** variant: baseline solid dielectric, smoother phase response (backs away from resonance).
- In case of print issues $\Rightarrow$ rhombus, hexagon-esque structure, etc.

The Physics: Form Birefringence + Resonance

Two mechanisms work together:

- **Form birefringence** sets the **anisotropy**.
  - Elliptical (not circular) cross-section.
  - Sub-wavelength feature $\Rightarrow$ different effective permittivity along major vs. minor axis: $n_{\parallel} \neq n_{\perp}$.
- **Reflection resonance** sets the **phase range**.
  - UC acts as de-tuned resonator.
  - Driving each axis through resonance swings the phase up to 360°.

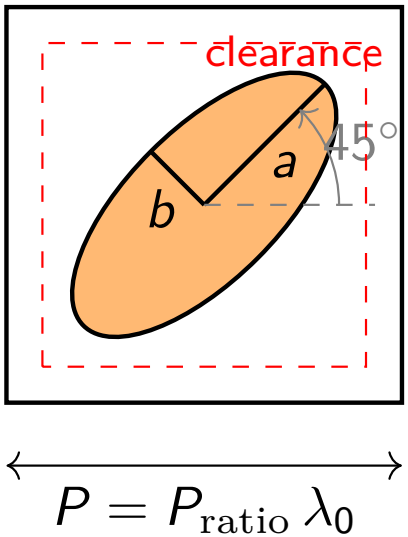
How the Two Mechanisms Couple

- Because the two axes have **different effective permittivities**, their resonances occur at **different pillar sizes** a.
- Form birefringence therefore **shifts the resonance location** between axes.
- The aspect ratio a/b tunes how far apart the two resonances sit, and hence the achievable 90° offset.

Design Tension

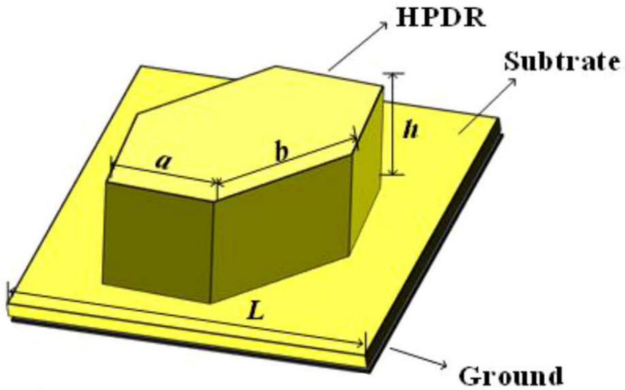
$\Delta\phi_x = 360°$ wants a *sharp* resonance;
a stable 90° offset to the other axis wants the two resonances *close together*. These pull in opposite directions.

Unit Cell Geometry — Parameters



- a, b — semi-major/minor axes of the element, $\frac{a}{b} = \texttt{ratio}$.
- P — cell footprint, set by `P_ratio`.
- Major axis rotated 45° relative to the array axes.
  - When illuminated by LP wave the incident wave can be seen as superposition of two LP waves aligned with principal axis.
- Clearance dictated by manufacturing technology – need to avoid thin walls, or bridging.

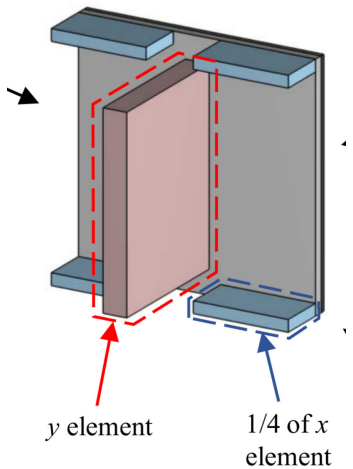
Unit Cell Design — Existing Studies



- Hexagonal-prism dielectric resonator[1]
  - Exhibits Form-Birefringence in x and y axis, profile is fixed.
  - Roughly equal-amplitude R_{xx} / R_{xy} and a **constant** 90° phase difference gives LP–CP conversion.
  - **Phase range from varying height**: 5.5 → 23 mm gives $> 360°$; the 90° offset stays constant with height.
  - Multiple modes at higher heights with high bandwidth?

[1] Jia-Wen Hao et al. "Linear-to-circular polarization conversion based on all-dielectric 3D-printed metasurface for the application of broadband circularly polarized reflectarray antenna". In: *Journal of Physics D: Applied Physics* 53.26 (2020), p. 265106.

Unit Cell Design — Existing Studies



- Five dielectric cuboids ($L \times W \times H$) in one cell[2].
  - The y element (L_y, W_y, H_y) reflects the y-polarized wave; the x element (comprised of 4 smaller blocks) reflects the x-polarized wave.
  - Phases tuned *independently* by H_x, H_y, giving two orthogonal LP waves that recombine into a CP wave.

[2] Qiao Cheng et al. "Dual Circularly Polarized 3-D Printed Broadband Dielectric Reflectarray With a Linearly Polarized Feed". In: *IEEE Transactions on Antennas and Propagation* 70.7 (2022), pp. 5393–5403.

How Our Cell Differs

- Both prior designs achieve the per-axis phase by varying **height** (Hao) or by **two separate height-tuned blocks** (Cheng).
- **Our proposal:** a **single element of uniform height**, where the two degrees of freedom come from **geometry changes in the x and y directions** rather than height.
  - Uniform height $\Rightarrow$ flat, mechanically robust surface.
  - No tall towers that block the wave reaching neighbouring cells – important for phase wraps on some zone boundaries.
- This variant does not appear to have been demonstrated before.

Challenges

- **Existence of the unit cell**
  - Can unit cell with large range phase range that maintains $\Delta\varphi = 90°$ exists?
  - At what frequencies are the properties reasonable?
- **Measurement**
  - Measuring permittivity of the material (most likely Zetamix 4.5 or 7.5)
  - Verifying properties of the unit cell – polarization, angle of incidence control.
    - Free-space quasi-optical approach
    - Parallel-plate waveguide – requires smaller sample, less known method.
- **Manufacturing**
  - Limited resolution of the printer – nozzle at min 0.2 mm.
  - Limited capabilities of the slicer – some studies set infill to e.g. 90 %, neglecting that for small features the slicer will print them solid.

Simulation Process — Overview

1. **Unit-cell model** in CST Microwave Studio (Floquet port, periodic boundaries).
2. **Excitation:** TE_{00} (x-pol) and TM_{00} (y-pol) modes, normal incidence.
3. Because the ellipse is rotated 45°, the x/y modes are not eigenpolarizations CST's output shows strong cross-coupling.
4. Recover the principal-axis phases by rotating the 2×2 reflection matrix by 45° in post-processing.
5. Sweep geometry, extract phase range and inter-axis offset.

Simulation Process — The Chain

- **CST**: parametric sweeps (ellipse size a, aspect ratio, permittivity, pillar height, cell size).
- **Export**: Touchstone `.s2p` per configuration + with VBA macro.
- **MATLAB**: load all runs, rotate each 2×2 matrix to the eigenbasis, extract
  - $|\phi_{\text{major}} - \phi_{\text{minor}}|$ (the 90° offset),
  - Find configuration with larger usable phase arc.
- Usable arc is dictated by $|S_{11}|$ and slope of phase change rate ($\leq 400^\circ/\text{mm}$)
  - Is ($\leq 400^\circ/\text{mm}$) too conservative? maybe – some studies use steeper slopes.
  - Lower frequencies will fair much better – UC is bigger, slope can be less steep.

Simulation Results

- Experimented mainly with pillar structure – replicated Beccarias's study[3]
- For initial exploratory study `ratio` is fixed to 1, $f_0 = 30$ GHz
- Sampling in a relatively low – just 25 samples per range.

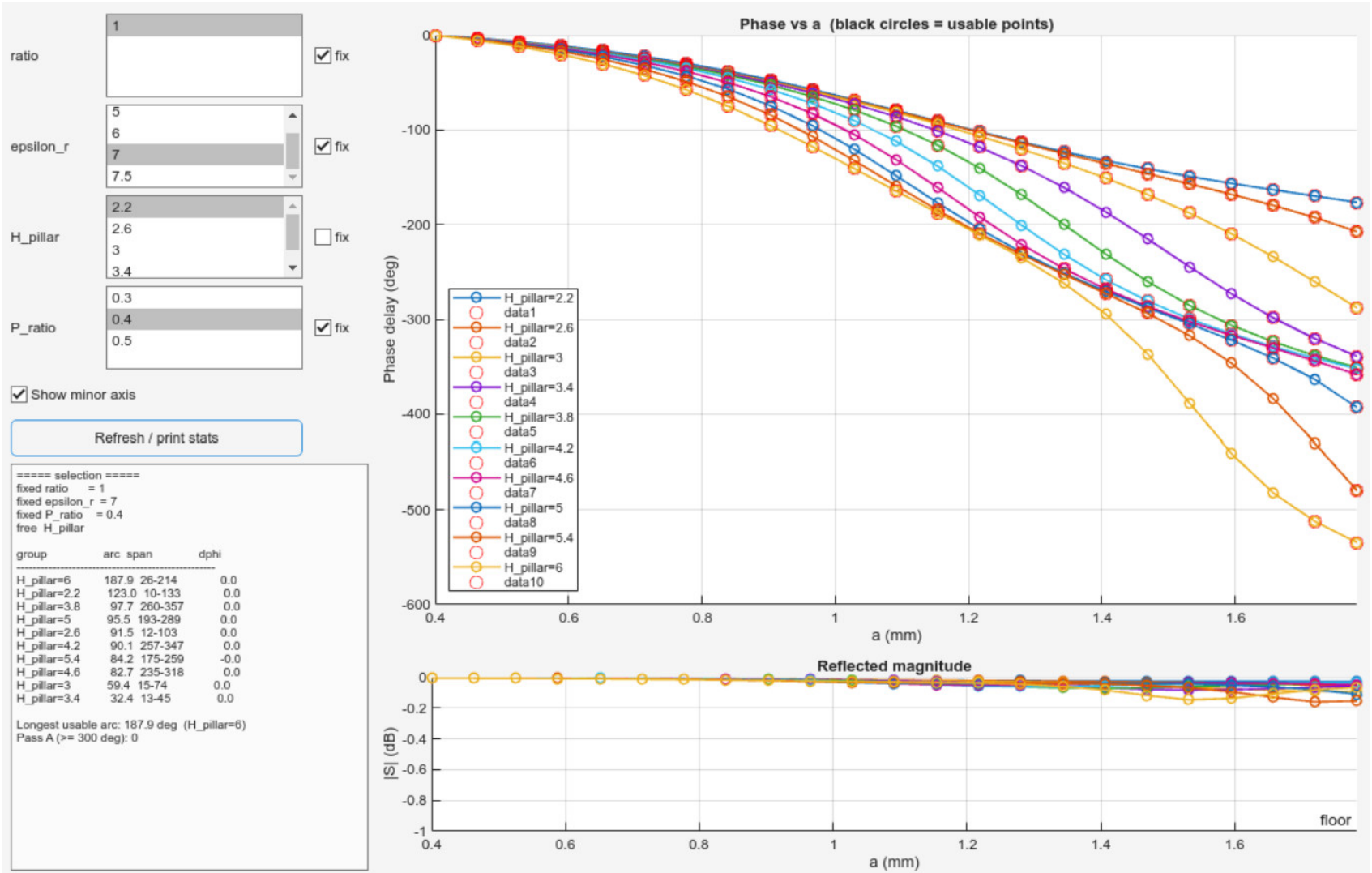
Phase range of cylindrical pillar

ε_r	P_{ratio}	H_{pillar} (mm)	$\Delta\varphi_x$ (°)
4.5	0.4	5	278.5
5	0.4	5	293.3
5	0.5	5	301.7
6	0.5	4.2	325.2

- $P_{\text{ratio}} = 0.5$, $\varepsilon_r > 6 \Rightarrow$ resonance within feasible range of a.

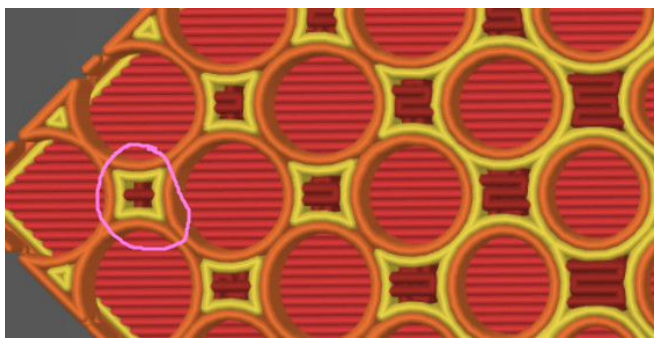
[3] Michele Beccaria et al. "Ka-Band Reflectarray with Cylindrical Dielectric Unit Cells: Optimized Additive Manufacturing and High-Permittivity Material Characterization". In: *Sensors* 25.17 (2025).

Simulation Results: Pillar, $P_{\text{ratio}} = 0.4$, $\varepsilon_r = 7$



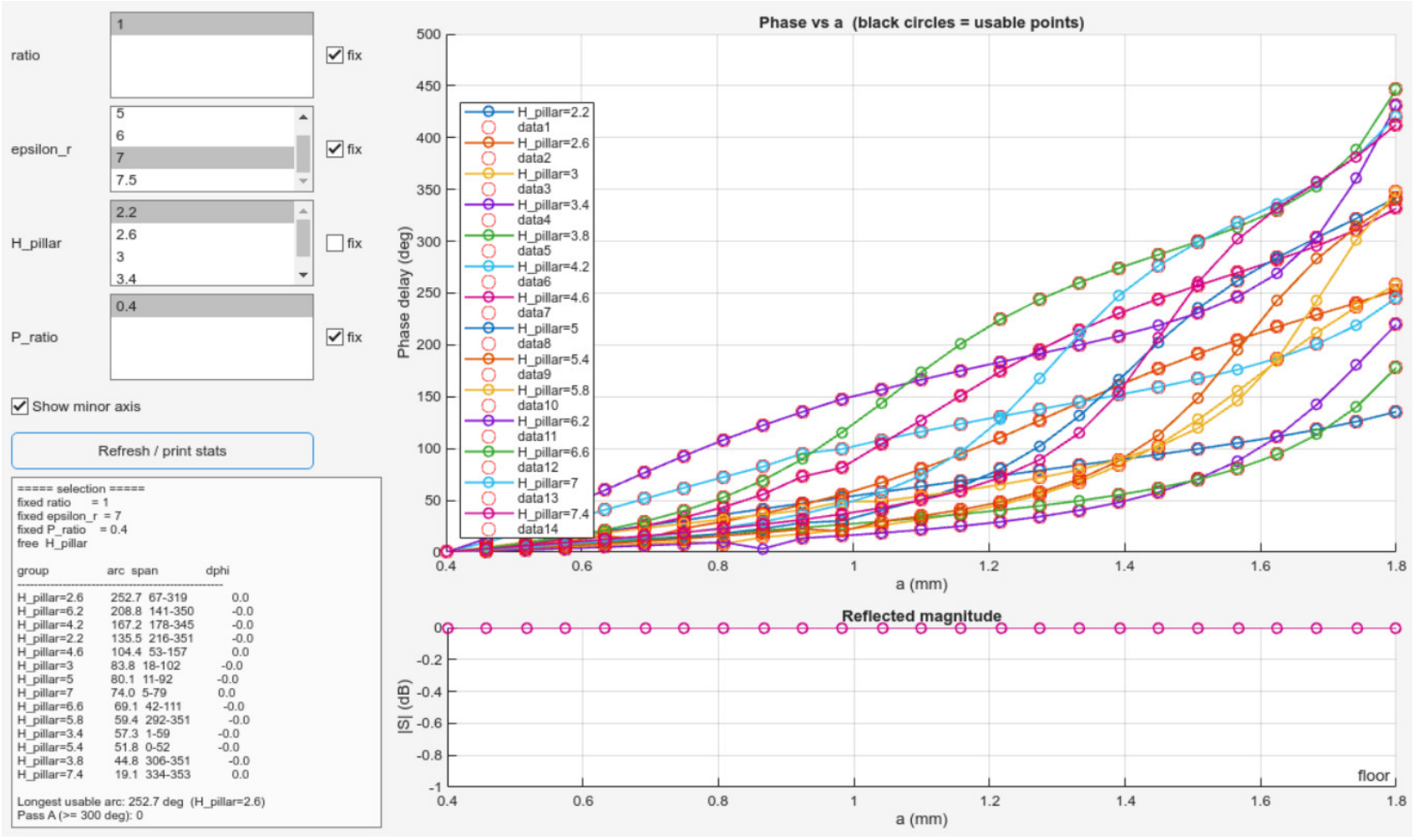
Simulation Results: Cylindrical Void

- Managed to run just $P_{\text{ratio}} = 0.4$ – sweep already took many hours.
- Highest range of 270.8° with $\varepsilon_r = 6$ and $H_{\text{pillar}} = 5$ mm
- Shape of the phase curves not too dissimilar, S_{11} seems better?
- Problems with 3D printing – pillars can use concentric infill.





Simulation Results: Void, $P_{\text{ratio}} = 0.4$, $\varepsilon_r = 7$



Why Not Reaching 360° Hurts Us More

- A **basic** reflectarray must hit any phase on the 0–360 ring to compensate the feed-to-element path. $\sim 300°$ is the accepted engineering threshold.
- Losing phase range raises a **phase error** at every element.
- In an **LP** array this error degrades **gain** and raises **sidelobes** – bad, but tolerable.

It is worse for LP–CP

- CP requires the two orthogonal components to stay at **equal amplitude and** 90° **apart** *at every element*.
- A missing phase range means the element cannot hit its required *offset*, not merely its required focus phase.
- Polarization-purity loss, not just a focusing loss.

Escalation Path if the Two Degrees of Freedom Fall Short

- If height + uniform geometry cannot deliver the range *and* a stable 90° offset:
  - add a **third degree of freedom** to the cell, or
  - adopt **multiple resonances**
- **More resonances:**
  - Each added resonance sharpens the phase slope, trading **bandwidth** for range.
  - Limited by printer resolution.
- **3 parameters:** (`a`, `ratio`, `H_pillar`) design is more complicated ⇒ MLP based surrogate model + NSGA-II to find optimal solutions if the behavior isn't nice.